

PRACTICAL - 9

Title:

Study of System Commands and Shell Script to Display System Configuration.

Requirements:

Linux Operating System (e.g., Ubuntu / Fedora / Kali Linux)

Bash Shell (Bourne Again Shell)

Terminal Emulator (default Linux terminal)

Theory:

A shell in Linux is a command-line interpreter that acts as an interface between the user and the operating system. It accepts user commands, interprets them, and executes them by interacting with the system kernel. The most commonly used shell in Linux is the Bash (Bourne Again Shell).

A shell script is a file that contains a sequence of commands written in a scripting language (usually Bash). Instead of typing commands one by one in the terminal, users can automate tasks by writing them into a script file and executing it. Shell scripting is widely used for system administration, automation, monitoring, and task scheduling.

Working of Shell Script

When a shell script is executed:

1. The script starts with a shebang (`#!/bin/bash`), which tells the system which interpreter to use.
2. The shell reads the script line by line.
3. Each command is executed in sequence.
4. Output is displayed on the terminal or redirected to files.

Environment Variables

Linux provides several environment variables that store important system information:

- `$HOME` → User's home directory
- `$PATH` → Directories where executable files are located
- `$SHELL` → Current shell being used

These variables are accessed using the `echo` command.

System Information Commands

Linux provides built-in utilities to retrieve system configuration details:

- User Information
 - `whoami` → Displays current user
 - `logname` → Shows login name
 - `who` → Lists logged-in users
- System Information
 - `uname -a` → Displays OS name, version, and kernel details
 - `/etc/os-release` → Detailed OS information
- Shell and Environment
 - `echo $SHELL` → Current shell
 - `echo $PATH` → Executable paths
- CPU Information
 - `lscpu` → Displays CPU architecture, cores, speed
- Memory Information
 - `free -h` → Shows RAM usage in human-readable format
- Disk Information
 - `df -h` → Displays disk space usage
 - `mount` → Lists mounted file systems
- Available Shells
 - `/etc/shells` → File containing list of valid shells

File Permissions in Shell Scripting

Before executing a shell script, it must be given execute permission using:

```
chmod +x filename.sh
```

Linux follows a permission model:

- `r` → Read
- `w` → Write
- `x` → Execute

Shell scripting is important because it:

- Automates repetitive tasks
- Reduces manual effort
- Helps in system monitoring and maintenance
- Enables quick retrieval of system information
- Improves productivity of system administrator

Program:

```
GNU nano 8.3 system_info.sh
#!/bin/bash

echo "===== SYSTEM CONFIGURATION DETAILS ====="

echo "Currently Logged User:"
whoami

echo "Login Name:"
logname

echo "Current Shell:"
echo $SHELL

echo "Home Directory:"
echo $HOME

echo "Path Settings:"
echo $PATH

echo "Current Working Directory:"
pwd

echo "Number of Logged Users:"
who | wc -l

echo "Operating System Name:"
uname -s

echo "OS Version:"
uname -v

echo "OS Release:"
uname -r

echo "Kernel Version:"
uname -a

echo "Available Shells:"
cat /etc/shells

echo "CPU Information (Type & Speed):"
lscpu

echo "Memory Information:"
free -h

echo "Hard Disk Information:"
df -h

echo "Mounted File Systems:"
mount

echo "===== END ====="

^G Help      ^O Write Out  ^F Where Is   ^X Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket M-B Previous
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-G Copy      ^B Where Was  M-F Next
```

Output:

```
rakhi@fedora:~$ nano system_info.sh
rakhi@fedora:~$ chmod +x system_info.sh
rakhi@fedora:~$ ./system_info.sh
===== SYSTEM CONFIGURATION DETAILS =====
Currently Logged User:
rakhi
Login Name:
rakhi
Current Shell:
/bin/bash
Home Directory:
/home/rakhi
Path Settings:
/home/rakhi/.local/bin:/home/rakhi/bin:/usr/local/bin:/usr/bin:/usr/local/sbin
Current Working Directory:
/home/rakhi
Number of Logged Users:
2
Operating System Name:
Linux
OS Version:
#1 SMP PREEMPT_DYNAMIC Mon Mar 24 19:53:37 UTC 2025
OS Release:
6.14.0-63.fc42.x86_64
Kernel Version:
Linux fedora 6.14.0-63.fc42.x86_64 #1 SMP PREEMPT_DYNAMIC Mon Mar 24 19:53:37 UTC 2025 x86_64 GNU/Linux
Available Shells:
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/usr/bin/tmux
/bin/tmux
CPU Information (Type & Speed):
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
On-line CPU(s) list:    0-15
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-13620H
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     10
Socket(s):              1
Stepping:               2
CPU(s) scaling MHz:     15%
CPU max MHz:            4900.0000
CPU min MHz:            400.0000
BogoMIPS:               5836.80
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge
                        mca cmov pat pse36 clflush dts acpi mmx fxsr sse ss
                        e2 ss ht tm pbe syscall nx pdpe1gb rdtscp lm constan
                        t_tsc art arch_perfmon pebs bts rep_good nopl xtopol
                        ogy nonstop_tsc cpuid aperfperf tsc_known_freq pni
                        pclmulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse
                        3 sdbg fma cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic
                        movbe popcnt tsc_deadline_timer aes xsave avx f16c r
                        dseed 3dcpuid 3dnowprefetch stibp clflushopt clflushwb
```

```

intel_pt sha_ni xsaveopt xsavec xgetbv1 xsaves split
_lock_detect user_shstk avx_vnni dtherm ida arat pln
pts hwp hwp_notify hwp_act_window hwp_epp hwp_pkg_r
eq hfi vnni umip pku ospke waitpkg gfni vaes vpclmul
dqg rdpid movdiri movdir64b fsrm md_clear serialize
arch_lbr ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization: VT-x
Caches (sum of all):
  L1d: 416 KiB (10 instances)
  L1i: 448 KiB (10 instances)
  L2: 9.5 MiB (7 instances)
  L3: 24 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Not affected
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via pr
    ctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user poin
    ter sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditio
    nal; RSB filling; PBRSSB-eIBRS SW sequence; BHI BHI_D
    IS_S
  Srbds: Not affected
  Tsx async abort: Not affected
Memory Information:
  total      used      free     shared buff/cache   available
Mem:    15Gi    3.2Gi    10Gi      734Mi     3.0Gi     12Gi
Swap:    8.0Gi         0B      8.0Gi
Hard Disk Information:
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p6  97G  4.3G   91G   5% /
devtmpfs        4.0M    0   4.0M   0% /dev
tmpfs           7.7G  96K  7.7G   1% /dev/shm
efivarfs        268K  151K  113K  58% /sys/firmware/efi/efivars
tmpfs           3.1G  2.3M  3.1G   1% /run
tmpfs           1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           7.7G  16K  7.7G   1% /tmp
/dev/nvme0n1p6  97G  4.3G   91G   5% /home
/dev/nvme0n1p5  974M  249M  658M  28% /boot
/dev/nvme0n1p1  256M   62M  195M  25% /boot/efi
tmpfs           1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           1.6G  212K  1.6G   1% /run/user/1000
Mounted File Systems:
/dev/nvme0n1p6 on / type btrfs (rw,relatime,seclabel,compress=zstd:1,ssd,discard=async,space_cache=v2,subvolid=256,subvol=/root)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4096k,nr_inodes=2000405,mode=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)

```

Result:

The shell script was successfully executed and displayed system configuration details such as user information, OS details, CPU, memory, disk usage, and mounted file systems.